11.5 User-defined Functions

11.5.1 Introduction

We can also define new functions to perform a specific task. These are called as **user-defined functions**. User-defined functions are created by the user. A function can optionally define input parameters that enable callers to pass arguments into the function. A function can also optionally return a value as output. Functions are useful for encapsulating common operations in a single reusable block, ideally with a name that clearly describes what the function does.

11.5.2 Function Definition

In C++, a function must be defined before it is used anywhere in the program. The general syntax of a function definition is:

Return_Data_Type Function_name(parameter list)

{

Body of the function

}

Note:

1. The Return_Data_Type is any valid data type of C++.
2. The Function_name is a user-defined identifier.
3. The parameter list, which is optional, is a list of parameters, i.e. a list of variables preceded by data types and separated by commas.
4. The body of the function comprises C++ statements that are required to perform the intended task of this function.

11.5.3 Function Prototype

C++ program can contain any number of functions. But, it must always have **only one main() function** to begin the program execution. We can write the definitions of functions in any order as we wish. We can define the main() function first and all other functions after that or we can define all the needed functions prior to main(). Like a variable declaration, a function must be declared before it is used in the program. The declaration statement may be given outside the main() function.

long fact (int, double)

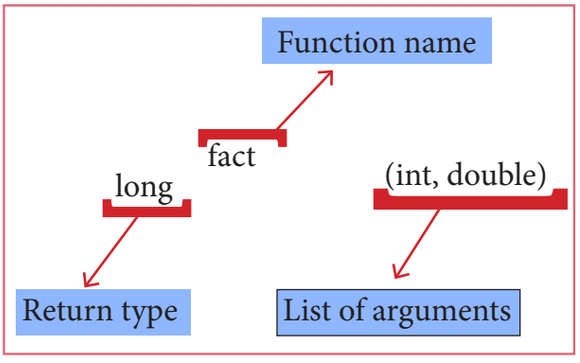

Figure 11.1

The prototype above provides the following information to the compiler:

- The return value of the function is of type long.
- **fact** is the name of the function.
- the function is called with two arguments:
  The first argument is of int **data** type.
  The second argument is of **double** data type.
  int display(int, int) // function prototype//

The above function prototype provides details about the return data type, name of the function and a list of formal parameters or arguments.

11.5.4 Use of void command

void type has two important purposes:

- To indicate the function does not return a value
- To declare a generic pointer.

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